

Post-selection-free measurement-altered criticality via statistical learning

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Measurements are useful for reshaping the universal properties of critical quantum states, but revealing the resulting measurement-altered criticality requires nonlinear probes of post-measurement ensembles that naively demand exponentially costly postselection. Here we prove that higher moments of conditioned observables can be estimated with controlled sample complexity. In transverse-field Ising model, we use machine-learning decoders from restricted functional classes to accurately reconstruct higher-moment one- and two-point functions, giving access to correlations characteristic of measurement-altered fixed points. Our approach provides a practical route to experimentally accessing measurement-altered criticality, including in Rydberg-array implementations.

Introduction. Measurements provide a powerful new control knob for quantum many-body dynamics. While projective measurements on the entire system erases quantum coherence and yields a trivial post-measurement state, measurements on a partial system can reshape the state of the unmeasured degrees of freedom in a nontrivial manner, offering an avenue for engineering quantum states and observing novel phenomena. A prominent example is the measurement-induced phase transition (MIPT), which occurs in the competition between unitary dynamics and random measurements applied to a trivial initial state. On the other hand, starting with a nontrivial initial state, partial/weak measurements can also non-perturbatively alter quantum correlations and entanglement. Recently, there have been extensive studies on measurement-altered criticality (MAC) in 1D quantum systems where some correlation power-law exponents can be non-perturbatively modified by measurements. Existing probes of MAC typically focus on post-selecting certain measurement outcomes or computing nonlinear functionals of the density matrix, both of which appear to demand an exponential number of snapshot samples and form an obstacle for experimental observation of MAC.

To circumvent this issue, previous works have proposed to use methods like quantum-classical correlations or physics-informed decoders to postprocess the experimental snapshot data. However, these methods relies on additional information of the underlying physics or has limited applicability. In this work, we propose using statistical learning models as a general and efficient decoder to extract the critical exponents of MAC from snapshot data only. We prove that the number of snapshots required to get converged estimation only scales with the model capacity and the desired accuracy. In critical Ising chains, we demonstrate numerically that polynomial-capacity models accurately recover MAC exponents from unpostselected measurement records, making measurement-altered criticality experimentally accessible.

(JA): Something like this could go naturally in the intro or discussion. The interest of this approach is thus twofold. Experimentally, it provides a post-selection-free route to extracting measurement-altered physics directly from snapshot data. Conceptually, it turns the nonlinear quantity $A_2[\mathcal{O}]$ into a controlled inference problem: machine learning enters not as a black-box add-on, but as a structured ansatz for the conditional response $\mathbf{s} \mapsto \langle \mathcal{O} \rangle_{\mathbf{s}}$.

[(YL) rydberg experiments]

General setup. We consider a chain of L qubits initialized into a quantum critical ground state ρ_0 whose universal long-distance properties are governed by a conformal field theory. Weak measurements are implemented by pairwise entangling the critical spins with L ancillary qubits and then projectively measuring the ancillas with outcome $\mathbf{s} = \{s_1, s_2, \dots, s_L\}$ arising with Born probability $p(\mathbf{s})$. Conditioned on $\mathbf{s}$, the system enters the non-unitarily modified state

$$\rho_{\mathbf{s}} = \frac{M_{\mathbf{s}} \rho_0 M_{\mathbf{s}}^\dagger}{p(\mathbf{s})}, \quad p(\mathbf{s}) = \text{Tr}(M_{\mathbf{s}} \rho_0 M_{\mathbf{s}}^\dagger) \quad (1)$$

with $M_{\mathbf{s}}$ the Kraus operator that imposes the information gleaned from the measurement. In field-theory language, the weak measurement generates a sub-dimensional perturbation—acting at a single imaginary-time slice but all positions—that we crucially assume is relevant for generic outcomes $\mathbf{s}$. In the thermodynamic limit such measurements *non-perturbatively* alter the critical spin chain’s properties for any typical $\mathbf{s}$ and any non-zero measurement strength.

Our goal is to harness data from either experiments or simulations to *efficiently* extract the resulting qualitative changes in measurement-altered observables judiciously averaged over $\mathbf{s}$. For any operator $\mathcal{O}$, a standard linear average of conditional expectation values reproduces the expectation value of the pre-measured state,

$$\sum_{\mathbf{s}} p(\mathbf{s}) \langle \mathcal{O} \rangle_{\mathbf{s}} = \langle \mathcal{O} \rangle, \quad (2)$$

and thus discards the outcome-to-outcome variations

that encode measurement-altered behavior of interest. Nontrivial measurement effects can, however, survive upon averaging higher moments and/or utilizing probabilities that differ from the Born probabilities $p(\mathbf{s})$ when summing over measurement outcomes. We therefore focus on generalized Born-averaged quantities of the form

$$A_q[\mathcal{O}_1, \mathcal{O}_2, \dots] \equiv \sum_{\mathbf{s}} p(\mathbf{s}) \langle \mathcal{O}_1 \rangle_{\mathbf{s}}^q \langle \mathcal{O}_2 \rangle_{\mathbf{s}}^q \dots \quad (3)$$

along with similar quantities uniformly weighted over measurement outcomes,

$$B_q[\mathcal{O}_1, \mathcal{O}_2, \dots] \equiv \frac{1}{2^L} \sum_{\mathbf{s}} \langle \mathcal{O}_1 \rangle_{\mathbf{s}}^q \langle \mathcal{O}_2 \rangle_{\mathbf{s}}^q \dots, \quad (4)$$

for some operators $\mathcal{O}_{1,2,\dots}$ and an exponent q specifying the moment of the expectation values. Such quantities are natural theoretical objects as they can be formulated in terms of (JA): *Is that only true for A? If so we can specify there here* [(YL) *B also has a replica representation*] replicated path integrals. From an experimental standpoint, however, their direct estimation appears extremely difficult at first glance. The conditional expectation values $\langle \mathcal{O} \rangle_{\mathbf{s}}$ require many repetitions with the same measurement record $\mathbf{s}$, but typical records occur with Born probability that decays exponentially with system size L ; moreover, nonstandard measurement-outcome probabilities used in Eq. (4) introduce an additional reweighting problem. Thus extracting A_q and B_q from raw data faces a severe post-selection bottleneck, motivating more efficient strategies that we turn to next.

Estimation of nonlinear observables. For an observable $\mathcal{O}$ diagonal in the measurement basis, here taken to be Z , brute-force estimation of the average $A_2[\mathcal{O}]$ evaluated with the Born probability $w(\mathbf{s}) = p(\mathbf{s})$ can proceed by repeating N protocol trials as follows. In a given trial $t = 1, \dots, N$, measure the ancillas with outcome $\mathbf{s}_t$ and then measure the system degrees of freedom in the Z basis with outcomes $\mathbf{z}_t$ —in turn yielding the observable value $\mathcal{O}(\mathbf{z}_t)$ for that ‘snapshot’. These data yield the direct $A_2[\mathcal{O}]$ estimator

$$\hat{A}_2^{\text{dir}}[\mathcal{O}] = \sum_{\mathbf{s}: N_{\mathbf{s}} > 0} \frac{N_{\mathbf{s}}}{N} \left[\frac{\sum_{t: \mathbf{s}_t = \mathbf{s}} \mathcal{O}(\mathbf{z}_t)}{N_{\mathbf{s}}} \right]^2, \quad (5)$$

where $N_{\mathbf{s}}$ denotes the number of snapshots in sector $\mathbf{s}$, the quantity in brackets estimates the conditional expectation value $\langle \mathcal{O} \rangle_{\mathbf{s}}$, and $N_{\mathbf{s}}/N$ estimates the Born probability $p(\mathbf{s})$. (Throughout hats denote estimators obtained from data averaged over trials.) Equation (5) is asymptotically correct, i.e., $\hat{A}_2^{\text{dir}}[\mathcal{O}] \rightarrow A_2[\mathcal{O}]$ at $N \rightarrow \infty$, yet can perform very poorly with a finite number of trials: Appendix A shows that the expected error cannot be made small unless $N \gtrsim O(e^{\alpha L})$ for some non-universal constant $\alpha > 0$ due to the post-selection problem highlighted above.

The same nonlinear quantity nevertheless admits a post-selection-free variational representation obtainable from the inequality

$$\sum_{\mathbf{s}} p(\mathbf{s}) [\langle \mathcal{O} \rangle_{\mathbf{s}} - f(\mathbf{s})]^2 \geq 0. \quad (6)$$

Rearranging yields the variational ansatz

$$A_2^{\text{var}}[\mathcal{O}] \equiv \sum_{\mathbf{s}} p(\mathbf{s}) [2f(\mathbf{s}) \langle \mathcal{O} \rangle_{\mathbf{s}} - f(\mathbf{s})^2] \leq A_2[\mathcal{O}] \quad (7)$$

that involves a standard Born average over the *linear* conditional expectation weighted by the ‘decoder’ function $f(\mathbf{s})$. The inequalities above are saturated for $f(\mathbf{s}) = \langle \mathcal{O} \rangle_{\mathbf{s}}$. Variationally estimating $A_2[\mathcal{O}]$ is therefore equivalent to learning the conditional response $\mathbf{s} \mapsto \langle \mathcal{O} \rangle_{\mathbf{s}}$ from snapshot data—which, crucially, need not require exponentially many trials.

In practice, we choose a hypothesis class $\mathcal{F}$ and minimize the empirical square loss

$$\hat{\mathcal{L}}_N(f) = \frac{1}{N} \sum_{t=1}^N [\mathcal{O}(\mathbf{z}_t) - f(\mathbf{s}_t)]^2 \quad (8)$$

over $f \in \mathcal{F}$. Denoting the learned decoder by $\hat{f}$, we arrive at the variational estimator

$$\hat{A}_2^{\text{var}}[\mathcal{O}] = \frac{1}{N} \sum_{t=1}^N [2\mathcal{O}(\mathbf{z}_t) \hat{f}(\mathbf{s}_t) - \hat{f}(\mathbf{s}_t)^2]. \quad (9)$$

Statistical-learning theory then implies that

$$N = O\left(\frac{\text{Pdim}(\mathcal{F}) \log \text{Pdim}(\mathcal{F}) + \log(1/\delta)}{\epsilon^2}\right) \quad (10)$$

snapshots suffice to guarantee that $\hat{A}_2^{\text{var}}[\mathcal{O}]$ is within ϵ of the true value $A_2^{\text{var}}[\mathcal{O}]$ with probability $1 - \delta$. Here $\text{Pdim}(\mathcal{F})$ is the pseudo-dimension of the decoder class, which roughly counts the number of free parameters in $\mathcal{F}$ (see Appendix A); for example, a linear decoder $f(\mathbf{s}) = c_0 + \sum_{j=1}^L c_j s_j$ with free parameters $c_0, \dots, c_L$ has $\text{Pdim}(\mathcal{F}) = L + 1$. The cost for high-fidelity estimation of $A_2^{\text{var}}[\mathcal{O}]$ is therefore controlled by the decoder capacity rather than by the exponentially many possible ancilla strings.

Minimizing the empirical square loss over *unrestricted* functions of $\mathbf{s}$ gives $\hat{f}(\mathbf{s}) = N_{\mathbf{s}}^{-1} \sum_{t: \mathbf{s}_t = \mathbf{s}} \mathcal{O}(\mathbf{z}_t)$. Substituting this optimum into $\hat{A}_2^{\text{var}}[\mathcal{O}]$ yields the estimator $\hat{A}_2^{\text{dir}}[\mathcal{O}]$, which again requires exponentially many trials to converge to $A_2[\mathcal{O}]$. Constraining the hypothesis class in a physically motivated way—e.g., imposing locality and multiscale structure in $\mathcal{F}$ —trades possible approximation bias for the ability to estimate nonlinear observables with controlled, manageable complexity. Given a restricted choice of function class, convergence of $\hat{A}_2^{\text{var}}[\mathcal{O}]$

to $A_2[\mathcal{O}]$ is not guaranteed. Below we nevertheless validate the performance by benchmarking the learned decoders against conditional expectation values and the resulting nonlinear observables computed using matrix product state (MPS) techniques.

Once learned and validated to give $f(\mathbf{s}) \approx \langle \mathcal{O} \rangle_{\mathbf{s}}$ with desired accuracy, the decoder can be leveraged to estimate various other nonlinear averages. For example, the approximate q -th moment reads

$$A_q[\mathcal{O}] = \sum_{\mathbf{s}} p(\mathbf{s}) \langle \mathcal{O} \rangle_{\mathbf{s}}^q \approx \frac{1}{N} \sum_{t=1}^N \hat{f}(\mathbf{s}_t)^q. \quad (11)$$

Given site-resolved decoders $f_j(\mathbf{s}) \approx \langle \mathcal{O}_j \rangle_{\mathbf{s}}$ and $f_k(\mathbf{s}) \approx \langle \mathcal{O}_k \rangle_{\mathbf{s}}$, disconnected conditioned correlators can be similarly estimated as

$$A_1[\mathcal{O}_j, \mathcal{O}_k] = \sum_{\mathbf{s}} p(\mathbf{s}) \langle \mathcal{O}_j \rangle_{\mathbf{s}} \langle \mathcal{O}_k \rangle_{\mathbf{s}} \approx \frac{1}{N} \sum_{i=1}^N f_j(\mathbf{s}_i) f_k(\mathbf{s}_i). \quad (12)$$

(JA): Awkward notational thing that I don't know how to address: previously we defined A_q in terms of a general weight w that may or may not be the Born distribution. Consistency would then suggest that we call both (11) and (12) $A_1[\mathcal{O}_j, \mathcal{O}_k]$, which is not a good idea. Any ideas for how to fix this? The same viewpoint allows efficient measurement-outcome averages with weights $w(\mathbf{s})$ differing from the Born distribution. In particular, with a flat outcome distribution $w(\mathbf{s}) = 1/2^L$, the uniform-outcome correlator

$$B[\mathcal{O}_j, \mathcal{O}_k] \equiv \frac{1}{2^L} \sum_{\mathbf{s}} \langle \mathcal{O}_j \rangle_{\mathbf{s}} \langle \mathcal{O}_k \rangle_{\mathbf{s}} \approx \frac{1}{N} \sum_{\mathbf{s}'} f_j(\mathbf{s}') f_k(\mathbf{s}') \quad (13)$$

can be approximated by evaluating the learned decoders on N uniformly sampled synthetic records, $\mathbf{s}' \sim \text{Uniform}\{\pm 1\}^L$. Accuracy of this observable depends on the decoder error under the uniform measure, rather than merely on the Born-weighted training loss. The uniform-outcome results below therefore provide a useful out-of-distribution benchmark of the learned conditional-response function.

Decoders for one-point functions. We train three types of machine-learning decoders for the one-point function in the measurement basis, $f_j(\mathbf{s}) \approx \langle Z_j \rangle_{\mathbf{s}}$. The simplest models are linear and logistic regressions (see Fig. 1(a,b)),

$$f_j^{\text{lin}}(\mathbf{s}) = c_0 + \sum_{k=1}^L c_k s_k, \quad f_j^{\text{log}}(\mathbf{s}) = \tanh \left(c_0 + \sum_{k=1}^L c_k s_k \right). \quad (14)$$

These models have $\text{Pdim}(\mathcal{F}) = O(L)$ and thus require only $O(L)$ snapshots. For a periodic chain, this number can be further reduced to $O(1)$ due to translational symmetry. Specifically, the linear model is expected to capture the leading dependence on $\mathbf{s}$ in the perturbative

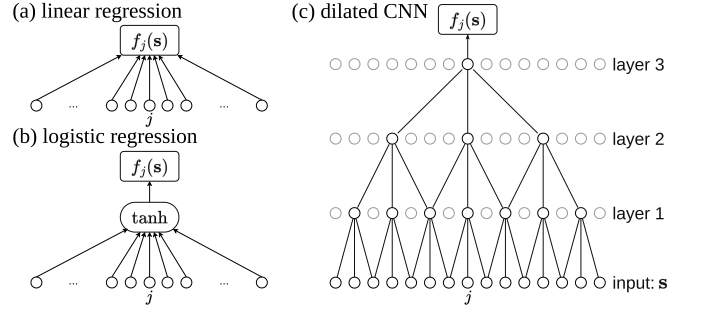


FIG. 1. Schematic comparison of the three architectures used for one-point-function decoders. Circles denote site-resolved measurement outcomes or hidden neurons.

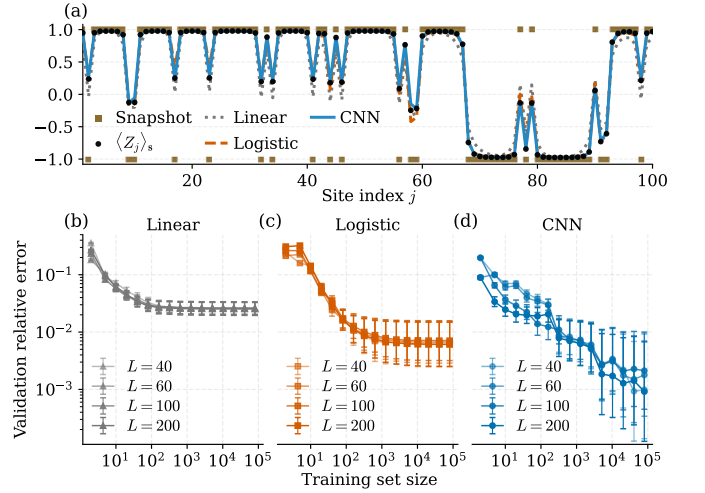


FIG. 2. Decoder benchmark for $\langle Z_j \rangle_{\mathbf{s}}$. (a): a representative record for $L = 100$, $\beta = 0.5$, comparing exact MPS conditional means (black dots) with linear, logistic and dilated CNN predictions. (b,c,d): relative validation error versus training-set size for $\beta = 0.5$ and several L . Errors saturate with far fewer than exponentially many samples.

regime: $\langle Z_j \rangle_{\mathbf{s}} \approx 2\beta \sum_k s_k \langle Z_j Z_k \rangle_0$ for $\beta \ll 1$. We also train a translation-covariant dilated convolutional neural network (CNN) [1]. The neural network consists of $O(\log(L))$ convolution layers with dilation $2^0, 2^1, 2^2, \dots$, capturing multi-scale dependence on $\mathbf{s}$ (see Fig. 1(c)). This is reminiscent of coarse graining and multiscale entanglement renormalization ansatz [2]. The exact architecture is given in Appendix B.

All these three models are trained using 80000 i.i.d. snapshot pairs sampled from the joint distribution $p(\mathbf{z}, \mathbf{s})$ and tested using another 20000 snapshots. Figure 2 shows that all three decoders learn the conditioned magnetization from snapshots, and the dilated CNN gives the smallest validation error. The top panel illustrates an example of random outcome and shows that ML-decoded one-point functions are close to the true $\langle Z_j \rangle_{\mathbf{s}}$ computed from MPS. The bottom panels show the validation mean-squared error of three models as a function of training-set

size N , which saturates at $N \gtrsim O(10)$ for linear and logistic regression and decreases with N for CNN. This indicates that these decoders can reach high accuracy with far fewer than exponentially many samples. We leave the numerical details in Appendix B.

The Born-weighted measurement ensemble admits a useful replica-field-theory description that controls both the weak-measurement scaling of $A_2[Z_j]$ and the long-distance moment correlations. For a fixed measurement record, define $\mathcal{Z}(\mathbf{s}) = \langle e^{2\beta \sum_j s_j Z_j} \rangle_0$. For integer R , the replicated weight $\mathcal{Z}(\mathbf{s})^R$ is represented by R copies of the Ising CFT coupled to the same record, with $S_{\mathbf{s},R} = \sum_a S_{\text{Ising}}^{(a)} - 2\beta \int dx s(x) \sum_a \sigma_a(x, 0)$, where a is the replica index. The measurement outcome average is performed on the Boltzmann weight, $\sum_{\mathbf{s}} e^{-S_{\mathbf{s},R}}$, where $\sum_{\mathbf{s}} \equiv \prod_j \sum_{s_j = \pm 1}$. This gives the replicated action

$$S_R = \sum_{a=1}^R S_{\text{Ising}}^{(a)} - \int dx \log \left[2 \cosh \left(2\beta \sum_{a=1}^R \sigma_a(x, 0) \right) \right], \quad (15)$$

with physical continuation $R \rightarrow 1$ due to the Born weight.

Expanding Eq. (15) at small β generates a defect perturbation $-g \int dx \sum_{a < b} \sigma_a \sigma_b + \dots$ with $g \propto \beta^2$. Since this operator has defect scaling dimension $2\Delta_\sigma$, the RG eigenvalue is $y_g = 1 - 2\Delta_\sigma$, implying a crossover length $\xi(\beta) \sim g^{-1/y_g} \sim \beta^{-2/(1-2\Delta_\sigma)}$. For the Ising CFT $\Delta_\sigma = 1/8$, this perturbation is relevant. Thus the unmeasured Ising fixed point controls length scales below $\xi(\beta)$, whereas the long-distance behavior is governed by a strong-coupling, locked-replica fixed point.

We now use this crossover scale to analyze $A_2[Z_j]$. Expanding the conditional mean gives [(NM) Shouldn't we be more careful about β being small compared to inverse system size? Can we say $\beta \rightarrow 0$, fixed L instead of $\beta \ll 1$?][(YL) Yes I agree]

$$\langle Z_j \rangle_{\mathbf{s}} = \frac{\langle Z_j e^{2\beta \sum_k s_k Z_k} \rangle_0}{\langle e^{2\beta \sum_k s_k Z_k} \rangle_0} \xrightarrow[\text{fixed } L]{\beta \ll 1} 2\beta \sum_k s_k \langle Z_j Z_k \rangle_0 + O(\beta^3), \quad (16)$$

and thus

$$A_2[Z_j] = \sum_{\mathbf{s}} p(\mathbf{s}) \langle Z_j \rangle_{\mathbf{s}}^2 = 4\beta^2 \sum_k \langle Z_j Z_k \rangle_0^2 + O(\beta^4). \quad (17)$$

Using $\langle Z_j Z_{j+r} \rangle_0 \sim r^{-2\Delta_\sigma}$ yields the perturbative scaling $A_2[Z_j] \sim \beta^2 L^{1-4\Delta_\sigma}$ for $L \ll \xi(\beta)$, while for $L \gtrsim \xi(\beta)$ the sum is cut off at $|j - k| \sim \xi(\beta)$. Matching the two regimes gives

$$A_2[Z_j] = \beta^{\frac{4\Delta_\sigma}{1-2\Delta_\sigma}} C \left(L \beta^{\frac{2}{1-2\Delta_\sigma}} \right), \quad (18)$$

[(NM) Should we mention that in the general case, our analysis only holds when $\Delta_\sigma \leq 1/4$ even though the measurement is relevant for $\Delta_\sigma < 1/2$?] with $C(x) \sim x^{1-4\Delta_\sigma} = x^{1/2}$ for $x \ll 1$ and $C(x) \rightarrow \text{const.}$ for $x \gtrsim 1$. Figure 3 shows that all the learned decoders reproduce

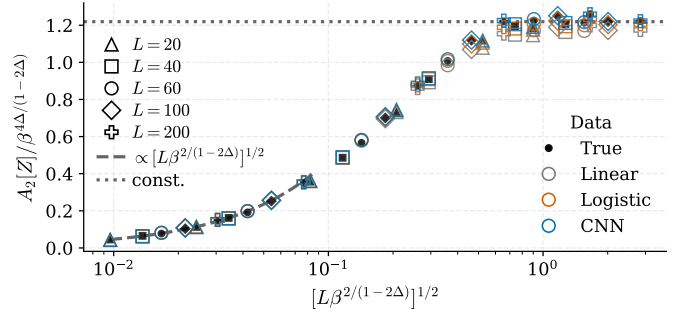


FIG. 3. Finite-size scaling collapse of $A_2[Z]$ for $\beta \leq 0.3$. The MPS reference and the learned linear, logistic and CNN estimators follow Eq. 18.

this finite-size collapse without exponential sample complexity.

Locked-replica fixed point and moment correlations. We now turn to the long-distance regime $r \gg \xi(\beta)$, where the same measurement-induced perturbation flows to the locked-replica fixed point. The higher moments of disconnected conditioned two-point functions provide direct probes of this fixed point. We consider $A_q[Z_j, Z_k] \equiv \sum_{\mathbf{s}} p(\mathbf{s}) \langle Z_j \rangle_{\mathbf{s}}^q \langle Z_k \rangle_{\mathbf{s}}^q$, with integer $q > 0$. For integer $R \geq 2q$, the replicated representation uses two disjoint groups of replicas: q spin insertions at site j and another q spin insertions at site k . Equivalently,

$$\mathcal{G}_{q,R}(j, k) = \left\langle \prod_{a=1}^q \sigma_a(j, 0) \prod_{a=q+1}^{2q} \sigma_a(k, 0) \right\rangle_{S_R}, \quad (19)$$

$$A_q[Z_j, Z_k] = \lim_{R \rightarrow 1} \mathcal{G}_{q,R}(j, k).$$

Replica permutation symmetry makes the particular choice of labels unimportant.

At the locked fixed point, the composite insertion in Eq. (19) behaves as the q th power of the single locked spin field. Hence odd q lies in the diagonal Ising-odd sector and has leading overlap with σ , while even q lies in the even sector and contains an identity component. The leading nontrivial even correction comes from the energy field ϵ . Therefore, on a ring with $d_L(r) = (L/\pi) \sin(\pi r/L)$,

$$A_q[Z_j, Z_{j+r}] \sim \begin{cases} b_q d_L(r)^{-1/4}, & q \text{ odd}, \\ c_q + a_q d_L(r)^{-2}, & q \text{ even}, \end{cases} \quad (20)$$

where a_q, b_q, c_q are nonuniversal. The powers follow from $\Delta_\sigma = 1/8$ and $\Delta_\epsilon = 1$ of the Ising CFT.

Figure 4 tests the odd/even structure predicted by the locked-replica picture. The odd moments with $q = 1, 3, 5, 7$ show a common algebraic decay consistent with the Ising-odd channel, $A_q[Z_j, Z_{j+r}] \sim d_L(r)^{-1/4}$. For even q , the most robust feature is the approach to a nonzero constant, as expected from the identity component in the even sector. The dashed curves show the

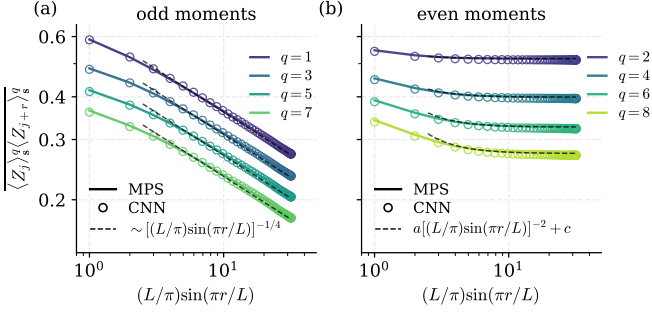


FIG. 4. Higher moments of disconnected conditioned two-point functions for the critical Ising chain at $L = 100$ and $\beta = 0.5$. The horizontal axis is the chord distance $d_L(r)$. MPS data are computed from 10000 randomly sampled conditioned post-measurement states, while CNN data use the learned decoder $f_j(\mathbf{s}) \simeq \langle Z_j \rangle_{\mathbf{s}}$ as a plug-in estimate (80000 training trials). Odd moments are consistent with the locked-replica prediction $d_L(r)^{-1/4}$, whereas even moments approach a constant; the dashed curves show the theory-motivated correction $c_q + a_q d_L(r)^{-2}$.

theory-motivated form $c_q + a_q d_L(r)^{-2}$, corresponding to the energy-field correction in Eq. (20). Given the limited scaling window and the small size of the subleading term, however, we regard the exponent -2 fit as a consistency check rather than an independent numerical determination. Higher moments emphasize outcomes with larger conditional response, so the agreement between MPS and CNN data provides a nontrivial check that the learned decoder captures the measurement-conditioned order-parameter field, not only its local variance.

An alternative way to estimate the scaling is to use a mean-field approximation of the nonlinear observable. Denoting $m_i(s) = \langle Z_i \rangle_{\mathbf{s}}$, for odd q we can write $m_i(s)^q = m_i(s)m_i(s)^{q-1}$, and replace the even factor by its typical value. Thus $m_i(s)^q$ flows to a linear order-parameter observable, $m_i(s)^q \sim A_q m_i(s)$, giving

$$\sum_{\mathbf{s}} p_{\mathbf{s}} m_i(s)^q m_j(s)^q \sim \sum_{\mathbf{s}} p_{\mathbf{s}} m_i(s) m_j(s) \sim \langle Z_i Z_j \rangle. \quad (21)$$

At the critical Ising point, this correlator decays as $|i - j|^{-1/4}$. For even q , instead, $m_i(s)^q$ is even under the Ising $\mathbb{Z}_2$ symmetry and its leading projection is the identity, so the averaged correlator approaches a constant. The first correction comes from the lowest nontrivial even operator, the Ising energy density ε , with scaling dimension 1. Therefore one expects

$$\sum_{\mathbf{s}} p_{\mathbf{s}} m_i(s)^q m_j(s)^q = C_0^{(q)} + C_1^{(q)} |i - j|^{-2} + \dots, \quad q \text{ even}, \quad (22)$$

where $C_0^{(q)}$ is the identity contribution and the $|i - j|^{-2}$ term is the leading connected correction.

[(NM) Can we say something about the higher moment connected correlators? We discussed previously that this

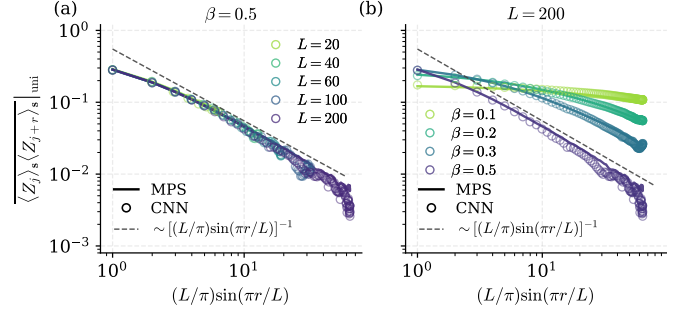


FIG. 5. Uniform-outcome disconnected conditioned correlations, $B(j, j+r) = 2^{-L} \sum_{\mathbf{s}} \langle Z_j \rangle_{\mathbf{s}} \langle Z_{j+r} \rangle_{\mathbf{s}}$. The horizontal axis is the chord distance $d_L(r) = (L/\pi) \sin(\pi r/L)$. MPS data are computed directly from conditioned post-measurement states, while CNN data use the learned decoder $f_j(\mathbf{s}) \simeq \langle Z_j \rangle_{\mathbf{s}}$ evaluated on uniformly sampled records. (a) Fixed $\beta = 0.5$ with varying L . (b) Fixed $L = 200$ with varying measurement strength. The dashed line is a guide to the free-boundary form $d_L(r)^{-1}$ expected for the zero-replica random-field defect. [(YL) TODO: update figure]

is where some “nontrivial” exponents may be hidden. Either we can say the numerics does not converge well enough or we can show whatever we have honestly. Unless I missed some discussion where we have more insight on these correlators. I worry a reader might look at our writing of the locked replica limit and go “all the measurement-altered observables are slight modifications of dephased critical Ising.”

Uniform-outcome correlations and the zero-replica limit. The Born-weighted observables above are tied to the physical $R \rightarrow 1$ continuation of the replicated measurement defect. We now consider a different ensemble of measurement records, the uniform-outcome average

$$B(j, k) \equiv \frac{1}{2^L} \sum_{\mathbf{s}} \langle Z_j \rangle_{\mathbf{s}} \langle Z_k \rangle_{\mathbf{s}}. \quad (23)$$

This quantity is not directly sampled by the Born distribution, but it is a useful theoretical companion because it removes the extra factor $p(\mathbf{s}) \propto \mathcal{Z}(\mathbf{s})$ that selected the $R \rightarrow 1$ limit. In practice, after training $f_j(\mathbf{s}) \simeq \langle Z_j \rangle_{\mathbf{s}}$, we estimate Eq. (23) by evaluating $f_j f_k$ on uniformly generated records.

Using $\mathcal{Z}_j(\mathbf{s}) = \langle Z_j e^{2\beta \sum_l s_l Z_l} \rangle_0$ and $\langle Z_j \rangle_{\mathbf{s}} = \mathcal{Z}_j(\mathbf{s})/\mathcal{Z}(\mathbf{s})$, the integer-replica object for $R \geq 2$ places the two insertions in different replicas. Analytic continuation gives

$$B(j, k) = \lim_{R \rightarrow 0} \langle \sigma_1(j, 0) \sigma_2(k, 0) \rangle_{S_R}, \quad (24)$$

with the same replicated action S_R as in Eq. (15). The denominator becomes $\sum_{\mathbf{s}} \mathcal{Z}(\mathbf{s})^R \rightarrow \sum_{\mathbf{s}} 1 = 2^L$, so the uniform average is the zero-replica counterpart of the Born-weighted problem [3].

It is useful to note a similar object — the same-replica correlator $B_{\text{sr}}(j, k) = 2^{-L} \sum_{\mathbf{s}} \langle Z_j Z_k \rangle_{\mathbf{s}} = \lim_{R \rightarrow 0} \langle \sigma_1(j, 0) \sigma_1(k, 0) \rangle_{S_R}$ which is addressed in the random-boundary-field CFT literature. In the Ising case, a quenched random magnetic field along a defect line is argued to flow to two decoupled half-plane Ising models with free boundary conditions [4, 5]. The boundary spin at the free Ising boundary has dimension $h_{\sigma}^{\text{free}} = 1/2$ [5, 6], leading to the familiar free-boundary decay $B_{\text{sr}}(j, j+r) \sim d_L(r)^{-1}$. The long-range contribution to the averaged correlator can be understood from the effective boundary problem produced by the replica construction. The replicas are glued together along the measured region, and this gluing line acts as a boundary defect. The large-distance decay is then controlled by the scaling dimension of the spin operator close to this defect. For the Ising chain, Ref. [7] argued that, for $R < 1$, this defect flows under RG to free boundary conditions. Thus, in the regime $R < 1$, the asymptotic behavior of the long-range contribution is expected to be controlled by the free-boundary fixed point. At this fixed point, the boundary spin has scaling dimension $h_{\sigma}^{\text{free}} = \frac{1}{2}$, which gives

$$B(j, j+r) \sim \tilde{b}_0(\beta) d_L(r)^{-2h_{\sigma}^{\text{free}}} = \tilde{b}_0(\beta) d_L(r)^{-1}. \quad (25)$$

This scaling form has been rigorously found in the replica geometry corresponding to $R = 1/2$ in Ref. [8]. Our numerical results suggest that the same free-boundary scaling also describes the continuation toward $R \rightarrow 0$, consistent with the RG picture of Ref. [7].

Our observable B is instead the inter-replica, disconnected counterpart. Thus the free-boundary result is not a direct theorem for Eq. (23). Nevertheless, the same zero-replica random-field flow suggests that its leading algebraic envelope should be governed by boundary spin insertions at the free-boundary fixed point. Provided the inter-replica amplitude generated by the random-field flow is nonzero, one expects

$$B(j, j+r) \sim \tilde{b}_0(\beta) d_L(r)^{-1}, \quad (26)$$

up to nonuniversal amplitudes and possible logarithmic corrections associated with random surface fields [4, 9]. We therefore regard Eq. (26) as a free-boundary-motivated prediction for the disconnected uniform average, to be checked numerically.

Figure 5 shows that B is consistent with this expectation. At fixed β , different system sizes follow a common $d_L(r)^{-1}$ trend over the accessible scaling window. At fixed L , the same trend appears for several measurement strengths, with stronger crossover effects at smaller β where the measurement length scale is larger. The dashed line is a guide with power-law exponent -1 . The agreement between MPS and CNN estimates indicates that the learned one-point-function decoder captures conditioned magnetization correlations even under the non-Born reweighting of measurement records.

Outlook. We have shown that nonlinear observables of measurement-conditioned states can be extracted from unpostselected snapshot pairs $(\mathbf{z}, \mathbf{s})$ by variational regression. For $A_2[\mathcal{O}]$, the optimal decoder is the conditional mean $f^*(\mathbf{s}) = \langle \mathcal{O} \rangle_{\mathbf{s}}$, so the sample cost is controlled by decoder capacity rather than by the exponentially many measurement records. In the critical Ising chain, translation-covariant decoders reproduce the finite-size scaling of $A_2[Z_j]$ and, used as plug-in estimators, capture higher-moment and uniform-outcome correlations associated with the $R \rightarrow 1$ locked-replica and $R \rightarrow 0$ random-defect limits.

The method is naturally suited to Rydberg arrays. Recent work proposed realizing MAC by projectively measuring subsets of atoms in Rydberg chains tuned to Ising or tricritical-Ising criticality [10], and experiments have begun resolving finite-size CFT spectra and dynamical response in such systems [11]. There the measured atoms provide $\mathbf{s}$ and the remaining readout provides $\mathbf{z}$, allowing order-parameter or energy-density decoders to use all shots and complement outcome-selective postselection.

[(YL) TODO: also mention Rushi and Ludwig paper]

There are potential interesting future directions. One is to replace purely classical post-processing by circuit decoders: the record $\mathbf{s}$ could control a shallow feed-forward unitary $U_{\theta}(\mathbf{s})$ before the final readout, or parametrized quantum circuits could be used as structured hybrid representations of the map $\mathbf{s} \mapsto \langle \mathcal{O} \rangle_{\mathbf{s}}$. Optimizing the same variational objective may then extend post-selection-free probes to non-diagonal observables, connected conditioned correlators, adaptive measurement bases, and other Rydberg-accessible universality classes. (JA): What about other quantum critical points including in higher dimensions?

END MATTER

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Appendix A: Sample-complexity bounds

In this appendix we give a self-contained derivation of the sample-complexity statements used in Sec. ???. We keep the notation of the main text: a single experimental run gives a system snapshot and an ancilla record, $(z, s) \sim p(z, s)$, and $O(z)$ is a diagonal observable of the system. We assume throughout the proof that

$$|O(z)| \leq 1. \quad (\text{A1})$$

This is the natural normalization for Pauli observables and their products. If instead $|O(z)| \leq B$, all statements below remain true with only changes of numerical constants by powers of B .

For a fixed measurement record s , define

$$m_O(s) := \mathbb{E}[O(z) \mid s] = \langle O \rangle_s. \quad (\text{A2})$$

The nonlinear observable considered in the main text is

$$A_2[O] := \mathbb{E}_s m_O(s)^2 = \sum_s p(s) \langle O \rangle_s^2. \quad (\text{A3})$$

All expectations below are taken with respect to the joint snapshot distribution $p(z, s)$ unless otherwise stated.

1. What is measured by $A_2[O]$?

The linear average of the conditional expectation does not retain outcome-to-outcome information,

$$\mathbb{E}_s m_O(s) = \mathbb{E}[\mathbb{E}[O(z) \mid s]] = \mathbb{E}[O(z)] = \langle O \rangle. \quad (\text{A4})$$

The second moment $A_2[O]$, by contrast, measures the fluctuations of the Bayes-optimal prediction of $O(z)$ from the ancilla record. Indeed,

$$A_2[O] - \langle O \rangle^2 = \text{Var}_s(m_O(s)). \quad (\text{A5})$$

The normalized quantity

$$\eta^2(O \mid s) := \frac{\text{Var}_s(m_O(s))}{\text{Var}(O(z))} \quad (\text{A6})$$

is the correlation ratio of $O(z)$ on the record s . It is the fraction of fluctuations of $O(z)$ explained by the best possible, generally nonlinear, predictor built from s . The law of total variance gives

$$\text{Var}(O(z)) = \mathbb{E}_s[\text{Var}(O(z) \mid s)] + \text{Var}_s(m_O(s)), \quad (\text{A7})$$

and therefore $0 \leq \eta^2(O \mid s) \leq 1$.

Relation to linear regression. If one restricts the predictor to be linear in the ancilla bits, the explained variance reduces to the usual linear-regression R^2 . For instance, if the bits s_j are treated as real random variables and O is centered, the best linear predictor has explained fraction

$$R^2 = \rho^T C^{-1} \rho, \quad (\text{A8})$$

where $\rho_j = \text{Corr}(O, s_j)$ and $C_{jk} = \text{Corr}(s_j, s_k)$, assuming C is nonsingular. Since η^2 optimizes over all measurable functions of s , one always has $R^2 \leq \eta^2(O \mid s)$. The learning approach in the main text should be viewed as an interpolation between these two extremes: the decoder class $\mathcal{F}$ is richer than a fixed analytic ansatz, but much smaller than the set of all functions on 2^L ancilla records.

Asymmetry and information. The correlation ratio is directional: in general $\eta^2(O \mid s) \neq \eta^2(s \mid O)$. It asks specifically how well the system observable $O(z)$ can be inferred from the measurement record s . It also gives a useful information-theoretic diagnostic. If $\rho_m(O; s)$ denotes the Hirschfeld–Gebelein–Rényi maximal correlation between the scalar random variable $O(z)$ and the record s , then

$$\eta^2(O \mid s) \leq \rho_m(O; s)^2. \quad (\text{A9})$$

Using the standard inequality $\rho_m(O; s)^2 \leq 1 - e^{-2I(O; s)}$, one obtains the lower bound

$$I(O; s) \geq -\frac{1}{2} \log[1 - \eta^2(O \mid s)]. \quad (\text{A10})$$

Thus a nonzero correlation ratio implies nonzero mutual information between $O(z)$ and the record s . In the present work we use $A_2[O]$ and $\eta^2(O \mid s)$ not primarily as information measures, but as experimentally accessible probes of the measurement-conditioned ensemble.

2. Direct estimator and exponential cost

Suppose we collect N independent snapshots $\{(z_i, s_i)\}_{i=1}^N$. For a fixed ancilla record s , define

$$N_s := \sum_{i=1}^N \mathbf{1}\{s_i = s\}, \quad S_s := \sum_{i:s_i=s} O(z_i), \quad (\text{A11})$$

$$\hat{m}_O(s) := \begin{cases} S_s/N_s, & N_s > 0, \\ 0, & N_s = 0. \end{cases} \quad (\text{A12})$$

The direct estimator in Sec. III is

$$\hat{A}_2^{\text{dir}}[O] = \sum_{s:N_s>0} \frac{N_s}{N} \hat{m}_O(s)^2 \quad (\text{A13})$$

$$= \frac{1}{N} \sum_s \frac{S_s^2}{N_s} \mathbf{1}\{N_s > 0\}. \quad (\text{A14})$$

This estimator explicitly estimates $\langle O \rangle_s$ separately in every post-selected sector that appears in the data.

Let

$$\sigma_O^2(s) := \text{Var}(O(z) \mid s). \quad (\text{A15})$$

Conditioned on $N_s = n > 0$, the n samples entering $\hat{m}_O(s)$ are i.i.d. draws from $p(z \mid s)$. Hence

$$\mathbb{E}[\hat{m}_O(s)^2 \mid N_s = n] = m_O(s)^2 + \frac{\sigma_O^2(s)}{n}. \quad (\text{A16})$$

Using Eq. (A16) inside Eq. (A14) gives

$$\mathbb{E} \hat{A}_2^{\text{dir}}[O] = A_2[O] + \frac{1}{N} \sum_s \sigma_O^2(s) \Pr(N_s > 0). \quad (\text{A17})$$

Thus the direct estimator is upward biased at finite N , although it becomes unbiased as $N \rightarrow \infty$ at fixed L .

Since $|O| \leq 1$ implies $\sigma_O^2(s) \leq 1$, Eq. (A17) gives the worst-case upper bound

$$0 \leq \mathbb{E} \hat{A}_2^{\text{dir}}[O] - A_2[O] \quad (\text{A18})$$

$$\leq \frac{1}{N} \sum_s \Pr(N_s > 0) = \frac{\mathbb{E} D_N}{N} \leq \frac{|\mathcal{S}|}{N}, \quad (\text{A19})$$

where $\mathcal{S}$ is the support of the ancilla record and

$$D_N := \#\{s \in \mathcal{S} : N_s > 0\} \quad (\text{A20})$$

is the number of distinct records observed in the sample. For an L -qubit ancilla, $|\mathcal{S}| \leq 2^L$. Thus making the bias term smaller than an additive accuracy ϵ requires $N \gtrsim 2^L/\epsilon$ in the worst case.

The exponential obstruction is the post-selection bias, not an anomalously large fluctuation around the mean. To see this, write

$$\hat{A}_2^{\text{dir}}[O] = \frac{1}{N} \sum_s a(N_s, S_s), \quad (\text{A21})$$

$$a(n, u) := \begin{cases} u^2/n, & n > 0, \\ 0, & n = 0. \end{cases} \quad (\text{A22})$$

Changing one snapshot affects at most two ancilla sectors. For $|O| \leq 1$, one has $|a(n, u) - a(n', u')| \leq 8$ for each affected sector under a single sample replacement, and therefore

$$\left| \hat{A}_2^{\text{dir}}(S) - \hat{A}_2^{\text{dir}}(S^{(i)}) \right| \leq \frac{16}{N}. \quad (\text{A23})$$

McDiarmid's inequality then gives

$$\Pr\left(\left| \hat{A}_2^{\text{dir}}[O] - \mathbb{E} \hat{A}_2^{\text{dir}}[O] \right| \geq \epsilon\right) \leq 2 \exp\left(-\frac{N\epsilon^2}{128}\right). \quad (\text{A24})$$

Equivalently, with probability at least $1 - \delta$,

$$\left| \hat{A}_2^{\text{dir}}[O] - A_2[O] \right| \leq \frac{1}{N} \sum_s \sigma_O^2(s) \Pr(N_s > 0) \quad (\text{A25})$$

$$+ \sqrt{\frac{128 \log(2/\delta)}{N}}. \quad (\text{A26})$$

The first term is the post-selection bias; the second is the ordinary concentration term.

One can also prove a lower bound showing that the exponential scaling is not just an artifact of Eq. (A19). Assume that the conditional variance is uniformly nonzero,

$$\sigma_O^2(s) \geq v > 0 \quad \text{for every } s \in \mathcal{S}. \quad (\text{A27})$$

Then Eq. (A17) implies

$$\mathbb{E} \hat{A}_2^{\text{dir}}[O] - A_2[O] \geq \frac{v}{N} \mathbb{E} D_N. \quad (\text{A28})$$

Furthermore,

$$D_N \sum_s N_s^2 \geq \left(\sum_s N_s \right)^2 = N^2. \quad (\text{A29})$$

Using Jensen's inequality and

$$\mathbb{E} \sum_s N_s^2 = N + N(N-1) \sum_s p(s)^2, \quad (\text{A30})$$

one obtains

$$\mathbb{E} D_N \geq \frac{N}{1 + (N-1) \sum_s p(s)^2}. \quad (\text{A31})$$

Therefore

$$\mathbb{E} \left| \hat{A}_2^{\text{dir}}[O] - A_2[O] \right| \geq \frac{v}{1 + (N-1) \sum_s p(s)^2}. \quad (\text{A32})$$

Let

$$H_2(s) := -\log \sum_s p(s)^2 \quad (\text{A33})$$

be the second Renyi entropy of the ancilla record. A necessary condition for the expected error of the direct estimator to be at most $\epsilon < v$ is

$$N \geq 1 + \left(\frac{v}{\epsilon} - 1 \right) e^{H_2(s)}. \quad (\text{A34})$$

Since $\sum_s p(s)^2 \leq p_{\max}$, where $p_{\max} := \max_s p(s)$, the weaker but simpler bound

$$N \geq 1 + \left(\frac{v}{\epsilon} - 1\right) \frac{1}{p_{\max}} \quad (\text{A35})$$

also follows. In the measurement protocols studied in the main text, the measurement record has exponentially large effective support, so $H_2(s) = \alpha L + o(L)$ for some $\alpha > 0$. Equation (A34) then gives $N \geq \Omega(e^{\alpha L})$ for the direct post-selection estimator.

3. Variational representation of $A_2[O]$

The post-selection-free approach starts from the identity

$$A_2[O] = \sup_f J(f), \quad (\text{A36})$$

$$J(f) := \mathbb{E}[2O(z)f(s) - f(s)^2], \quad (\text{A37})$$

where the supremum is over bounded measurable functions of the ancilla record. Conditioning on s gives

$$J(f) = \mathbb{E}_s[2m_O(s)f(s) - f(s)^2] \quad (\text{A38})$$

$$= \mathbb{E}_s m_O(s)^2 - \mathbb{E}_s[f(s) - m_O(s)]^2. \quad (\text{A39})$$

Thus the optimal decoder is the Bayes predictor

$$f^*(s) = m_O(s) = \langle O \rangle_s, \quad (\text{A40})$$

and the maximum value is $A_2[O]$.

Equivalently,

$$2O(z)f(s) - f(s)^2 = O(z)^2 - [O(z) - f(s)]^2. \quad (\text{A41})$$

Therefore maximizing $J(f)$ is the same as minimizing the mean-squared prediction loss

$$L(f) := \mathbb{E}[O(z) - f(s)]^2. \quad (\text{A42})$$

This is the precise sense in which the nonlinear observable $A_2[O]$ is converted into a supervised-learning problem.

In practice the decoder is restricted to a hypothesis class $\mathcal{F}$ of functions $f : \mathcal{S} \rightarrow [-1, 1]$. Define the best-in-class value

$$J_{\mathcal{F}}^* := \sup_{f \in \mathcal{F}} J(f) \quad (\text{A43})$$

$$= A_2[O] - \inf_{f \in \mathcal{F}} \mathbb{E}_s[f(s) - \langle O \rangle_s]^2. \quad (\text{A44})$$

The second term is the approximation error of the decoder class. Thus the learning method is not a no-free-lunch procedure: if the map $s \mapsto \langle O \rangle_s$ is completely unstructured, a small decoder class will not approximate it. The gain comes from using the physical structure of measurement-conditioned states, encoded either explicitly or through the inductive bias of the model, to choose a class $\mathcal{F}$ whose capacity is much smaller than 2^L .

4. Uniform convergence and sample complexity

Given data $\{(z_i, s_i)\}_{i=1}^N$, define the empirical objective

$$\hat{J}_N(f) := \frac{1}{N} \sum_{i=1}^N [2O(z_i)f(s_i) - f(s_i)^2] \quad (\text{A45})$$

and choose

$$\hat{f}_N \in \arg \max_{f \in \mathcal{F}} \hat{J}_N(f). \quad (\text{A46})$$

Equivalently, $\hat{f}_N$ minimizes the empirical mean-squared loss

$$\hat{L}_N(f) := \frac{1}{N} \sum_{i=1}^N [O(z_i) - f(s_i)]^2. \quad (\text{A47})$$

The estimator reported in Sec. III is

$$\hat{A}_{2,\mathcal{F}}[O] := \hat{J}_N(\hat{f}_N) \quad (\text{A48})$$

$$= \frac{1}{N} \sum_{i=1}^N [2O(z_i)\hat{f}_N(s_i) - \hat{f}_N(s_i)^2]. \quad (\text{A49})$$

Let

$$\mathcal{H}_{\mathcal{F}} := \{h_f(z, s) = 2O(z)f(s) - f(s)^2 : f \in \mathcal{F}\} \quad (\text{A50})$$

be the induced objective-function class, and let

$$d_{\text{obj}} := \text{Pdim}(\mathcal{H}_{\mathcal{F}}) \quad (\text{A51})$$

be its pseudo-dimension. The pseudo-dimension is the real-valued analogue of VC dimension: it is the largest number of inputs that can be threshold-shattered by functions in the class. For the decoder classes used in the main text, d_{obj} is of the same order as the pseudo-dimension of the decoder class itself. In scaling estimates we therefore write $d_{\text{obj}} = \Theta(\text{Pdim}(\mathcal{F}))$.

Because $|O| \leq 1$ and $|f| \leq 1$, every h_f lies in an interval of length at most 4. Standard uniform-convergence bounds for bounded real-valued function classes imply that, for a universal constant C , with probability at least $1 - \delta$,

$$\Delta_N := \sup_{f \in \mathcal{F}} |\hat{J}_N(f) - J(f)| \quad (\text{A52})$$

$$\leq C \left[\frac{d_{\text{obj}} \log(eN/d_{\text{obj}}) + \log(1/\delta)}{N} \right]^{1/2}, \quad (\text{A53})$$

for $N \geq d_{\text{obj}}$.

Let $f_{\mathcal{F}}^* \in \arg \max_{f \in \mathcal{F}} J(f)$. On the event where Eq. (A53) holds,

$$J(\hat{f}_N) \geq \hat{J}_N(\hat{f}_N) - \Delta_N \quad (\text{A54})$$

$$\geq \hat{J}_N(f_{\mathcal{F}}^*) - \Delta_N \quad (\text{A55})$$

$$\geq J(f_{\mathcal{F}}^*) - 2\Delta_N. \quad (\text{A56})$$

Thus the learned decoder has excess objective value bounded by

$$0 \leq J_{\mathcal{F}}^* - J(\hat{f}_N) \leq 2\Delta_N. \quad (\text{A57})$$

The empirical optimum itself is also close to the best-in-class value:

$$\hat{J}_N(\hat{f}_N) \geq \hat{J}_N(f_{\mathcal{F}}^*) \geq J_{\mathcal{F}}^* - \Delta_N, \quad (\text{A58})$$

$$\hat{J}_N(\hat{f}_N) \leq J(\hat{f}_N) + \Delta_N \leq J_{\mathcal{F}}^* + \Delta_N. \quad (\text{A59})$$

Therefore

$$\left| \hat{A}_{2,\mathcal{F}}[O] - J_{\mathcal{F}}^* \right| \leq \Delta_N. \quad (\text{A60})$$

Combining Eqs. (A44) and (A60) yields the decomposition

$$\left| \hat{A}_{2,\mathcal{F}}[O] - A_2[O] \right| \leq \Delta_N \quad (\text{A61})$$

$$+ \inf_{f \in \mathcal{F}} \mathbb{E}_s [f(s) - \langle O \rangle_s]^2. \quad (\text{A62})$$

The first term is the statistical estimation error, and the second term is the approximation error.

Solving Eq. (A53) for $\Delta_N \leq \epsilon$ gives

$$N = O\left(\frac{\text{Pdim}(\mathcal{F}) \log \text{Pdim}(\mathcal{F}) + \log(1/\delta)}{\epsilon^2}\right), \quad (\text{A63})$$

up to universal constants and mild logarithmic factors. This is the sample-complexity estimate quoted in Sec. III. The number of snapshots is controlled by the decoder capacity rather than by the number of possible ancilla strings.

Examples of capacities. For the linear decoder

$$f(s) = c_0 + \sum_{j=1}^L c_j s_j, \quad (\text{A64})$$

the pseudo-dimension is $L+1$ before imposing additional symmetries or bounds. If a translation-covariant ansatz with a fixed number of shared parameters is used, the parameter count, and hence the capacity entering the bound, can be reduced accordingly. For neural-network decoders, standard pseudo-dimension bounds scale polynomially in the number of trainable weights; a common bound is of the form

$$\text{Pdim}(\mathcal{F}) = O(W\ell \log W), \quad (\text{A65})$$

where W is the number of trainable parameters and ℓ is the number of layers. Thus a decoder whose number of parameters grows polynomially with system size has polynomial sample complexity. Conversely, if $\mathcal{F}$ is taken to be the set of all functions on L ancilla bits, then $\text{Pdim}(\mathcal{F}) = 2^L$, and Eq. (A63) reproduces the exponential post-selection scaling.

Training and evaluation on the same data. Equation (A60) is a uniform-convergence result, so it allows the same data to be used both to train $\hat{f}_N$ and to evaluate $\hat{J}_N(\hat{f}_N)$. A more conservative experimental protocol is to train $\hat{f}_N$ on one set of snapshots and evaluate Eq. (A49) on an independent validation set. Conditioned on the trained decoder, the validation estimate is an ordinary average of bounded random variables and is therefore controlled directly by Hoeffding's inequality. The scaling with the capacity of $\mathcal{F}$ enters only through the training step.

5. Plug-in estimates for other conditioned observables

The proof above is stated for $A_2[O]$ because the square loss gives the exact variational identity Eq. (A37). Once a decoder $f(s) \approx m_O(s)$ has been learned, however, the same function can be used as a representation of the conditioned ensemble.

Let

$$e_O(f) := \mathbb{E}_s [f(s) - m_O(s)]^2 \quad (\text{A66})$$

be the mean-squared decoder error. If $|f| \leq 1$ and $|m_O| \leq 1$, then for integer $q \geq 1$,

$$|f(s)^q - m_O(s)^q| \leq q|f(s) - m_O(s)|. \quad (\text{A67})$$

Therefore

$$|\mathbb{E}_s f(s)^q - A_q[O]| \leq q\sqrt{e_O(f)}. \quad (\text{A68})$$

The same bound applies to absolute moments $\mathbb{E}_s |m_O(s)|^q$ for $q \geq 1$ after replacing f^q by $|f|^q$.

For two observables $O_a(z)$ and $O_b(z)$, define $m_a(s) = \langle O_a \rangle_s$ and $m_b(s) = \langle O_b \rangle_s$. Suppose f_a and f_b are bounded decoders with errors e_a and e_b in the sense of Eq. (A66). Then the disconnected conditioned correlator obeys

$$|\mathbb{E}_s f_a(s) f_b(s) - \mathbb{E}_s m_a(s) m_b(s)| \quad (\text{A69})$$

$$\leq \sqrt{e_a} + \sqrt{e_b}. \quad (\text{A70})$$

This justifies the plug-in estimates used in the main text, such as

$$\sum_s p(s) \langle O_a \rangle_s \langle O_b \rangle_s \approx \frac{1}{N} \sum_{i=1}^N f_a(s_i) f_b(s_i), \quad (\text{A71})$$

with the usual additional sampling error if the right-hand side is evaluated on a finite validation set.

6. Connected conditional correlators

The same variational idea can be applied to connected conditional correlators. Let $O_a(z)$ and $O_b(z)$ be two

bounded observables and define

$$C_{ab}(s) := \langle O_a O_b \rangle_s - \langle O_a \rangle_s \langle O_b \rangle_s, \quad (\text{A72})$$

$$B_{ab} := \mathbb{E}_s C_{ab}(s)^2. \quad (\text{A73})$$

If the one-point conditional means $m_a(s) = \langle O_a \rangle_s$ and $m_b(s) = \langle O_b \rangle_s$ were known exactly, then the residual variable

$$R_{ab}(z, s) := [O_a(z) - m_a(s)] [O_b(z) - m_b(s)] \quad (\text{A74})$$

satisfies

$$\mathbb{E}[R_{ab}(z, s) | s] = C_{ab}(s). \quad (\text{A75})$$

Hence

$$B_{ab} = \sup_g \mathbb{E} [2R_{ab}(z, s)g(s) - g(s)^2], \quad (\text{A76})$$

with optimizer $g^*(s) = C_{ab}(s)$. This is the same square-loss identity as before, applied to the bounded target R_{ab} .

With finite data, a clean implementation uses sample splitting. First, use one subset of snapshots to learn $\hat{m}_a(s)$ and $\hat{m}_b(s)$. Second, on an independent subset, form

$$\hat{R}_{ab,i} := [O_a(z_i) - \hat{m}_a(s_i)] [O_b(z_i) - \hat{m}_b(s_i)]. \quad (\text{A77})$$

Third, learn a decoder $g(s)$ for $\hat{R}_{ab}$ by maximizing

$$\frac{1}{N} \sum_i \left[2\hat{R}_{ab,i}g(s_i) - g(s_i)^2 \right]. \quad (\text{A78})$$

Conditioned on the first subset, the functions $\hat{m}_a$ and $\hat{m}_b$ are fixed on the second subset. If

$$e_a(s) := \hat{m}_a(s) - m_a(s), \quad e_b(s) := \hat{m}_b(s) - m_b(s), \quad (\text{A79})$$

then the conditional mean of the residual target used in the second stage is

$$\mathbb{E}[\hat{R}_{ab}(z, s) | s] = C_{ab}(s) + e_a(s)e_b(s). \quad (\text{A80})$$

Thus the first-stage bias in the connected target is quadratic in the one-point decoder errors. The sample complexity of the last step is again controlled by the pseudo-dimension of the decoder class for g , with additional approximation terms inherited from the first-stage decoders. This construction is not needed for the main numerical results, but it shows that genuinely connected conditioned observables can also be treated without explicit sector-by-sector postselection.

7. Other losses and related conditional quantities

The square loss is special because it learns the conditional mean, which is the quantity entering $A_2[O]$. Other losses learn other conditional summaries and can be useful diagnostics of the same measurement record.

Absolute loss and conditional medians. For a real random variable X , the minimizer of $\mathbb{E}[|X - a|]$ over constants a is a median of X . Therefore minimizing

$$\mathbb{E}[|O(z) - f(s)|] \quad (\text{A81})$$

over functions f learns the conditional median of $O(z)$ given s , rather than the conditional mean. For binary observables $O(z) \in \{\pm 1\}$, this conditional median is the majority-vote predictor $\text{sign } m_O(s)$, up to ties.

For binary $O(z) \in \{\pm 1\}$, the first absolute moment of the conditional mean can be obtained from the optimal classification accuracy. For a classifier $h(s) \in \{\pm 1\}$,

$$\text{Acc}(h) := \Pr[h(s) = O(z)] \quad (\text{A82})$$

$$= \mathbb{E} \left[\frac{1 + O(z)h(s)}{2} \right] = \frac{1 + \mathbb{E}_s[m_O(s)h(s)]}{2}. \quad (\text{A83})$$

Maximizing over all classifiers gives $h^*(s) = \text{sign } m_O(s)$ and

$$\mathbb{E}_s|m_O(s)| = 2 \text{Acc}^* - 1. \quad (\text{A84})$$

Restricting h to a classifier class gives a post-selection-free estimator whose sample complexity is governed by the VC dimension, or an analogous capacity measure, of that class.

Cross entropy and conditional entropy. If a coarse-grained system label $\chi(z)$ takes values in $\{1, \dots, K\}$, a probabilistic decoder $q_\theta(k | s)$ trained by cross entropy learns the conditional distribution of χ given s . For any candidate conditional distribution $q(k | s)$,

$$\mathbb{E}[-\log q(\chi(z) | s)] = H(\chi | s) \quad (\text{A85})$$

$$+ \mathbb{E}_s D_{\text{KL}}(p(\cdot | s) \| q(\cdot | s)) \quad (\text{A86})$$

$$\geq H(\chi | s), \quad (\text{A87})$$

with equality for $q(k | s) = p(\chi = k | s)$. Thus the optimum cross entropy is the conditional entropy, and $I(\chi; s) = H(\chi) - H(\chi | s)$ gives the mutual information between the label and the measurement record. Unlike the bounded square-loss setting used for Eq. (A63), the logarithmic loss is unbounded unless the predicted probabilities are clipped away from zero. With such clipping, analogous uniform-convergence bounds can be applied, with constants depending on the clipping scale.

8. Summary

The direct estimator of $A_2[O]$ is asymptotically correct but suffers a post-selection bias proportional to the number of distinct ancilla sectors that must be populated. Under broad conditions this requires $N = \Omega(e^{\alpha L})$ snapshots. The learning estimator avoids this by using the variational identity

$$A_2[O] = \sup_f \mathbb{E} [2O(z)f(s) - f(s)^2], \quad (\text{A88})$$

which turns the problem into regression of $O(z)$ on s . For a decoder class $\mathcal{F}$, the number of snapshots needed to estimate the best-in-class value to additive error ϵ with failure probability δ scales as

$$N = O\left(\frac{\text{Pdim}(\mathcal{F}) \log \text{Pdim}(\mathcal{F}) + \log(1/\delta)}{\epsilon^2}\right), \quad (\text{A89})$$

up to universal constants and mild logarithmic factors, plus the approximation error of representing $s \mapsto \langle O \rangle_s$ within $\mathcal{F}$. This is the sense in which the cost is controlled by decoder complexity rather than by the exponentially many possible measurement records.

Appendix B: Decoder architectures and numerical training protocol

This appendix gives the architectural and training details for the decoders used in Sec. ???. We keep the notation of the main text. A snapshot is a pair (z, s) sampled from the joint Born distribution $p(z, s)$, where $z_j, s_j \in \{\pm 1\}$ are the system and ancilla outcomes. For the one-point function we train a decoder

$$f_j(s) \simeq m_j(s), \quad m_j(s) := \mathbb{E}[z_j | s] = \langle Z_j \rangle_s, \quad (\text{B1})$$

for every site j . The decoders are trained only on raw snapshot pairs (z, s) . The exact conditional values $m_j(s)$ computed by MPS are not used as training labels; they are used only for the reference curves and for the error estimates described below.

1. Implementation conventions for the hyperparameters

Several names used below are implementation flags in the numerical code rather than standard machine-learning terminology. We define them here before listing the numerical choices for each decoder.

The option `fit_intercept` specifies whether an affine bias b is included in the linear or logistic map. In the notation of the main text, this is the constant term c_0 in Eqs. ?? and ??. The option `pbcs_shared` enforces the periodic translation symmetry at the level of the trainable parameters. Instead of fitting independent coefficients $\{b_j, w_r^{(j)}\}$ for each output site j , it fits one reference-site decoder g with a single set of coefficients $\{b, w_r\}$ and predicts every other site by translating the ancilla record to the reference frame,

$$f_j(s) = g(T_{-j}s), \quad (\text{B2})$$

where T_{-j} shifts the target site j to a fixed reference position. Equivalently, with periodic indices understood modulo L , the linear shared-PBC model has the form

$f_j(s) = b + \sum_r w_r s_{j+r}$. Thus the parameters are translation invariant in relative coordinates, while the site-resolved output is translation covariant, or equivariant: $f_{j+a}(T_a s) = f_j(s)$. If `pbcs_shared` is not used, one instead fits independent site-wise models with independent parameter vectors. Finally, `augment_z2` denotes augmentation by the global Ising symmetry: for each training example (s, z) we also include $(-s, -z)$. This enforces, or at least strongly biases the fit toward, the physically expected odd response $f_j(-s) = -f_j(s)$.

For the logistic regression, `solver` is the numerical optimizer, `max_iter` is its maximum number of iterations, and C is the inverse strength of the standard ℓ_2 penalty in the logistic objective. Thus larger C corresponds to weaker regularization. For the CNN, the parameters listed below are the number of hidden channels, convolution kernel size, dilation base, learning rate, batch size, and number of training epochs.

2. Data sets and translational averaging

For each pair (L, β) used in the figures, we generated independent snapshot pairs $(z^{(a)}, s^{(a)})$ from the MPS representation of the measured critical Ising chain. In the convergence and validation plots, the data set contains

$$N_{\text{tot}} = 10^5 \quad (\text{B3})$$

raw snapshot pairs. We used a fixed random split with

$$N_{\text{val}} = 2 \times 10^4, \quad N_{\text{train}}^{\text{max}} = 8 \times 10^4. \quad (\text{B4})$$

The validation set is disjoint from the training set. When the training-size dependence is plotted, the training set of size N_{train} is chosen as a nested subset of the fixed training pool, with

$$N_{\text{train}} \in \{2, 5, 10, 20, 40, 80, 160, 320, 640, 1280, 2560, 5120, 10240, 20480, 40960, 80000\}. \quad (\text{B5})$$

Thus different points in a convergence curve are evaluated on the same validation set.

The chain is periodic in the numerical experiments. For the linear and logistic decoders, we use the shared-PBC construction defined in Eq. (B2). Equivalently, the shared training set is

$$\mathcal{D}_{\text{sh}} = \{(T_{-j}s^{(a)}, z_j^{(a)}) : a = 1, \dots, N_{\text{train}}, j = 1, \dots, L\}. \quad (\text{B6})$$

This realizes the statement in Sec. ??? that translation symmetry can reduce the number of independent parameters relative to fitting an unrelated model at every site. The shared linear and logistic maps still have $L + 1$ coefficients, but a single snapshot supplies labels for all

L translated sites, so translation averaging reduces the number of independent snapshot records needed in practice. For the convolutional neural network, translation covariance is built into the architecture by weight sharing and circular padding, so no explicit shifted data set is required.

We also use the global Ising symmetry in the data augmentation step. Whenever a training snapshot (s, z) is used, we include its symmetry partner

$$(s, z) \mapsto (-s, -z). \quad (\text{B7})$$

For the convex linear and logistic objectives this makes the optimal intercept zero up to numerical tolerance. For the neural network it acts as symmetry augmentation and strongly biases the learned decoder toward $f_j(-s) = -f_j(s)$.

3. Linear decoder

The linear decoder is the translationally shared version of Eq. ?? of the main text. With periodic indices understood modulo L , it can be written as

$$f_j^{\text{lin}}(s) = \text{clip}_{[-1,1]} \left(b + \sum_{r=0}^{L-1} w_r s_{j+r} \right). \quad (\text{B8})$$

The clipping is applied only at prediction time to keep the output in the physical range $[-1, 1]$. The fit itself is ordinary least-squares regression on the augmented shared data set in Eq. (B6). Numerically, the linear runs include the affine bias b , impose the shared-PBC translation-covariant form, use the global $\mathbb{Z}_2$ data augmentation, and apply no explicit regularization. No separate cyclic-shift augmentation is used beyond the shared-PBC pooling. In the code notation defined above, this is `fit_intercept=True`, `psc_shared=True`, and `augment_z2=True`. The number of trainable parameters in the shared PBC implementation is $L + 1$ before the $\mathbb{Z}_2$ symmetry sets the intercept to zero. Without the shared-PBC constraint, a separate linear model for every site would have $L(L + 1)$ parameters.

4. Logistic decoder

The logistic decoder uses the same shared-PBC feature map as the linear decoder, but it learns the conditional probability of a positive system outcome. For $u = T_{-j}s$, define

$$p_+(u) = \frac{1}{1 + \exp[-(b + \sum_{r=0}^{L-1} w_r u_r)]}. \quad (\text{B9})$$

The magnetization prediction is

$$f_j^{\text{log}}(s) = 2p_+(T_{-j}s) - 1 = \tanh \left[\frac{1}{2} \left(b + \sum_{r=0}^{L-1} w_r s_{j+r} \right) \right]. \quad (\text{B10})$$

After a harmless rescaling of b and w_r , this is the bounded logistic form shown in Eq. ?? of the main text.

The model is trained by binary cross entropy on the labels $(z_j + 1)/2 \in \{0, 1\}$. The Bayes optimum of this classification problem is $p_+(s) = \Pr(z_j = +1 | s)$, and therefore the corresponding magnetization is $2p_+(s) - 1 = \langle Z_j \rangle_s$. Thus the logistic loss and the squared loss have the same target conditional mean, although they weight finite-sample errors differently.

Numerically, the logistic runs use the same shared-PBC and $\mathbb{Z}_2$ -augmented data convention as the linear runs, with an affine bias included and the shared-PBC pooling. The optimizer is `lbfgs`, the maximum number of optimizer iterations is 400, and $C = 1.0$ is the inverse strength of the standard ℓ_2 regularization in the logistic objective. In the code notation defined above, this is `solver=lbfgs`, `C=1.0`, `max_iter=400`, `fit_intercept=True`, `psc_shared=True`, and `augment_z2=True`. The shared model has $L + 1$ trainable parameters before the symmetry constraint on the intercept.

5. Dilated convolutional neural network

The neural-network decoder is a one-dimensional dilated convolutional neural network with periodic padding. Unlike the linear and logistic baselines, it outputs all sites simultaneously,

$$f_\theta(s) = (f_{\theta,1}(s), \dots, f_{\theta,L}(s)) \in [-1, 1]^L. \quad (\text{B11})$$

The input is viewed as a one-channel signal of length L . The architecture is:

1. A circular convolutional stem with one input channel, $C = 64$ hidden channels, kernel size $K = 3$, dilation 1, followed by the SiLU activation $\text{SiLU}(x) = x/(1 + e^{-x})$.
2. A stack of B residual dilated convolutional blocks. The ℓ th block, with $\ell = 0, \dots, B - 1$, has dilation

$$d_\ell = 2^\ell, \quad (\text{B12})$$

$C = 64$ input and output channels, kernel size $K = 3$, and circular padding. If $h^{(\ell)}$ is the hidden representation entering the block, the update is

$$h^{(\ell+1)} = h^{(\ell)} + \text{SiLU} \left(\text{Conv}_{K=3, d_\ell}^{\text{circ}} h^{(\ell)} \right). \quad (\text{B13})$$

3. A pointwise readout convolution with kernel size 1, mapping the $C = 64$ hidden channels to one output channel at each site, followed by a final tanh nonlinearity,

$$f_\theta(s) = \tanh\left(\text{Conv}_{K=1} h^{(B)}\right). \quad (\text{B14})$$

The final tanh enforces the physical bound $f_{\theta,j}(s) \in [-1, 1]$.

The number of residual blocks is chosen automatically so that the receptive field covers the periodic chain. With the stem included, the receptive-field length after B residual blocks is

$$R_B = 3 + 2 \sum_{\ell=0}^{B-1} 2^\ell = 2^{B+1} + 1. \quad (\text{B15})$$

We choose the smallest B such that $R_B \geq L$. Thus $B = O(\log L)$, and with fixed channel number the number of trainable weights also scales as $O(\log L)$. For the hyperparameters used here, the parameter count is

$$\begin{aligned} W_{\text{CNN}} &= 64(3+1) + B(64^2 \cdot 3 + 64) + (64+1) \\ &= 321 + 12352B. \end{aligned} \quad (\text{B16})$$

For the system sizes used in the figures this gives the values in Table I.

TABLE I. Number of dilated residual blocks and trainable parameters for the CNN with $C = 64$, $K = 3$, and dilation base 2.

L	B	receptive field R_B	trainable parameters
20	4	33	49729
40	5	65	62081
60	5	65	62081
100	6	129	74433
200	7	257	86785

The CNN is trained as a Bernoulli decoder for all sites. Its magnetization output is converted to a probability

$$p_{\theta,j}(+ | s) = \frac{f_{\theta,j}(s) + 1}{2}, \quad (\text{B17})$$

and the target is $y_j = (z_j + 1)/2$. The training loss is the site-averaged binary cross entropy

$$\begin{aligned} \mathcal{L}_{\text{BCE}}(\theta) &= -\frac{1}{N_{\text{train}}L} \sum_{a=1}^{N_{\text{train}}} \sum_{j=1}^L [y_j^{(a)} \log p_{\theta,j}(+ | s^{(a)}) \\ &\quad + (1 - y_j^{(a)}) \log(1 - p_{\theta,j}(+ | s^{(a)}))]. \end{aligned} \quad (\text{B18})$$

with the probabilities clipped to the interval $[10^{-6}, 1 - 10^{-6}]$ for numerical stability. The optimizer is Adam

with learning rate 10^{-3} , batch size 256, and 20 epochs. No dropout or weight decay is used.

Hyperparameter summary. The numerical settings used for the three decoders in Sec. IV are as follows. The implementation-specific names in parentheses are defined in Sec. B 1.

Linear decoder: Ordinary least-squares regression; affine bias included (`fit_intercept`); translationally shared PBC model (`pbc_shared`); global Ising-symmetry augmentation (`augment_z2`); no explicit regularization.

Logistic decoder: Bernoulli logistic regression for $\text{Pr}(z_j = +1 | s)$; affine bias included; translationally shared PBC model; global Ising-symmetry augmentation; `lbfgs` optimizer; inverse ℓ_2 regularization strength $C = 1.0$; maximum iteration count 400.

Dilated CNN: Circular one-dimensional residual CNN with hidden channels $C = 64$, kernel size $K = 3$, dilation base 2, and automatic number B of residual blocks chosen from Eq. (B15); SiLU activations; final tanh output; binary-cross-entropy loss; Adam optimizer; learning rate 10^{-3} ; batch size 256; 20 epochs; global Ising-symmetry augmentation; no dropout and no weight decay.

6. Validation estimator and reference MPS error bars

The validation set is used to estimate the variational objective in Sec. ?? without reusing the training data. For a trained model $\mathcal{M}$, define the site-averaged validation mean-squared error

$$\widehat{\mathcal{L}}_{\text{val}}^{\mathcal{M}} := \frac{1}{N_{\text{val}}L} \sum_{a=1}^{N_{\text{val}}} \sum_{j=1}^L [z_j^{(a)} - f_j^{\mathcal{M}}(s^{(a)})]^2. \quad (\text{B19})$$

Because $z_j^2 = 1$, the corresponding validation estimate of the site-averaged second moment is

$$\widehat{A}_{2,\mathcal{M}}^{\text{val}} = 1 - \widehat{\mathcal{L}}_{\text{val}}^{\mathcal{M}}, \quad (\text{B20})$$

where

$$\overline{A}_2[Z] := \frac{1}{L} \sum_{j=1}^L A_2[Z_j] = \frac{1}{L} \sum_{j=1}^L \mathbb{E}_s \langle Z_j \rangle_s^2. \quad (\text{B21})$$

Indeed, if $f_j^{\mathcal{M}}(s) = \langle Z_j \rangle_s$ is the Bayes decoder, then

$$\mathbb{E} [(z_j - f_j^{\mathcal{M}}(s))^2] = 1 - A_2[Z_j]. \quad (\text{B22})$$

Thus the irreducible validation MSE is $1 - \overline{A}_2[Z]$.

For the MPS reference, we independently sample

$$N_{\text{MPS}} = 10^4 \quad (\text{B23})$$

measurement records from the Born distribution and compute the true conditional expectation values $m_j^{\text{MPS}}(s) = \langle Z_j \rangle_s$ by MPS contraction. The reference value is

$$\hat{A}_2^{\text{MPS}} = \frac{1}{N_{\text{MPS}}L} \sum_{a=1}^{N_{\text{MPS}}} \sum_{j=1}^L \left[m_j^{\text{MPS}}(s^{(a)}) \right]^2, \quad (\text{B24})$$

or, equivalently, the reference irreducible MSE is

$$\hat{\mathcal{L}}_{\text{MPS}} = 1 - \hat{A}_2^{\text{MPS}}. \quad (\text{B25})$$

The single-record comparison shown in Sec. ?? uses the same object $m_j^{\text{MPS}}(s)$, but for one displayed measurement record s rather than for the average in Eq. (B24).

The validation statistical error is estimated over snapshots rather than over individual sites, so correlations among different sites in the same snapshot are not treated as independent samples. Define

$$\ell_a^{\mathcal{M}} := \frac{1}{L} \sum_{j=1}^L \left[z_j^{(a)} - f_j^{\mathcal{M}}(s^{(a)}) \right]^2. \quad (\text{B26})$$

Then

$$\text{SE}_{\text{val}}^{\mathcal{M}} = \sqrt{\frac{1}{N_{\text{val}}(N_{\text{val}} - 1)} \sum_{a=1}^{N_{\text{val}}} \left(\ell_a^{\mathcal{M}} - \hat{\mathcal{L}}_{\text{val}}^{\mathcal{M}} \right)^2} \quad (\text{B27})$$

is the standard error of the validation MSE. The MPS standard error is computed analogously from the N_{MPS} per-record values

$$\ell_a^{\text{MPS}} := 1 - \frac{1}{L} \sum_{j=1}^L \left[m_j^{\text{MPS}}(s^{(a)}) \right]^2. \quad (\text{B28})$$

The relative errors in the main text are errors in the inferred second moment, normalized by the MPS reference value,

$$\epsilon_{\text{rel}}^{\mathcal{M}} := \frac{\left| \hat{A}_{2,\mathcal{M}}^{\text{val}} - \hat{A}_2^{\text{MPS}} \right|}{\hat{A}_2^{\text{MPS}}} = \frac{\left| \hat{\mathcal{L}}_{\text{val}}^{\mathcal{M}} - \hat{\mathcal{L}}_{\text{MPS}} \right|}{1 - \hat{\mathcal{L}}_{\text{MPS}}}. \quad (\text{B29})$$

The plotted error bars combine the finite- N_{val} statistical uncertainty of the ML validation estimate and the finite- N_{MPS} uncertainty of the MPS reference. In the convergence plots we used the conservative quadrature estimate

$$\text{SE}(\epsilon_{\text{rel}}^{\mathcal{M}}) \simeq \frac{\sqrt{(\text{SE}_{\text{val}}^{\mathcal{M}})^2 + 2(\text{SE}_{\text{MPS}})^2}}{\hat{A}_2^{\text{MPS}}}. \quad (\text{B30})$$

The factor of two is a conservative way of accounting for the fact that the MPS estimate enters both the numerator and the normalization in Eq. (B29). If one linearizes only the numerator, the same formula appears without the second copy of SE_{MPS} ; in the present data the difference is smaller than the scale of the trends discussed in the main text. These error bars quantify statistical uncertainty from finite MPS and validation samples.